Part (a): Highest Demand Rate in 2017

SQL

```
SELECT t.timestamp, c.demand
FROM CarSharing_df c
JOIN time t ON c.id = t.id
WHERE t.timestamp LIKE '2017%'
ORDER BY c.demand DESC
LIMIT 1;
```

Part (b): Highest & Lowest Average Demand by Weekday, Month, and Season

SQL

```
SELECT t.weekday_name, t.month_name, t.season, AVG(c.demand) AS avg_demand
FROM CarSharing_df c
JOIN time t ON c.id = t.id
WHERE t.timestamp LIKE '2017%'
GROUP BY t.weekday_name, t.month_name, t.season
ORDER BY avg_demand DESC;
```

Part (c): Average Demand by Hour for the Selected Weekday

SQL

```
SELECT t.hour, AVG(c.demand) AS avg_demand
FROM CarSharing_df c
JOIN time t ON c.id = t.id
WHERE t.timestamp LIKE '2017%' AND t.weekday_name = 'Selected_Weekday'
GROUP BY t.hour
ORDER BY avg_demand DESC;
```

Part (d): Weather & Temperature Impact on Demand / Monthly Wind & Humidity

SQL

```
-- Average demand by temperature category and weather condition
SELECT tmp.temp_category, w.weather, AVG(c.demand) AS avg_demand
FROM CarSharing_df c
JOIN weather w ON c.weather_code = w.weather_code
JOIN temperature tmp ON c.temp_code = tmp.temp_code
JOIN time t ON c.id = t.id
WHERE t.timestamp LIKE '2017%'
GROUP BY tmp.temp_category, w.weather
ORDER BY avg_demand DESC;
```

```
-- Monthly wind speed statistics for 2017
SELECT t.month_name, AVG(tmp.windspeed) AS avg_wind, MAX(tmp.windspeed) AS
max_wind, MIN(tmp.windspeed) AS min_wind
FROM CarSharing_df c
JOIN temperature tmp ON c.temp_code = tmp.temp_code
JOIN time t ON c.id = t.id
WHERE t.timestamp LIKE '2017%'
GROUP BY t.month_name;

-- Monthly humidity statistics for 2017
SELECT t.month_name, AVG(tmp.humidity) AS avg_humidity, MAX(tmp.humidity) AS
max_humidity, MIN(tmp.humidity) AS min_humidity
FROM CarSharing_df c
JOIN temperature tmp ON c.temp_code = tmp.temp_code
JOIN time t ON c.id = t.id
WHERE t.timestamp LIKE '2017%'
GROUP BY t.month_name;
```

Part (e): Comparison for the Month with Highest Average Demand

```
SQL
SELECT t.month_name, AVG(c.demand) AS avg_demand
FROM CarSharing_df c
JOIN time t ON c.id = t.id
WHERE t.timestamp LIKE '2017%'
GROUP BY t.month_name
ORDER BY avg_demand DESC
LIMIT 1;

SELECT t.month_name, tmp.temp_category, w.weather,
    AVG(c.demand) AS avg_demand,
    AVG(tmp.windspeed) AS avg_wind,
    AVG(tmp.humidity) AS avg_humidity
FROM CarSharing_df c
JOIN time t ON c.id = t.id
JOIN weather w ON c.weather_code = w.weather_code
JOIN temperature tmp ON c.temp_code = tmp.temp_code
WHERE t.timestamp LIKE '2017%'
GROUP BY t.month_name, tmp.temp_category, w.weather
ORDER BY avg_demand DESC
LIMIT 5;
```